

Practice 1: Closed-loop control of a neural circuit (Part I)

1-Build a Hindmarsh Rose neuron using the HR chaotic model with the following parameters:

$$\frac{dx_i(t)}{dt} = y_i(t) + 3x_i^2(t) - x_i^3(t) - z_i(t) + e_i$$

$$\frac{dy_i(t)}{dt} = 1 - 5x_i^2(t) - y_i(t)$$

$$\frac{1}{\mu} \frac{dz_i(t)}{dt} = -z_i(t) + S [x_i(t) + 1.6]$$

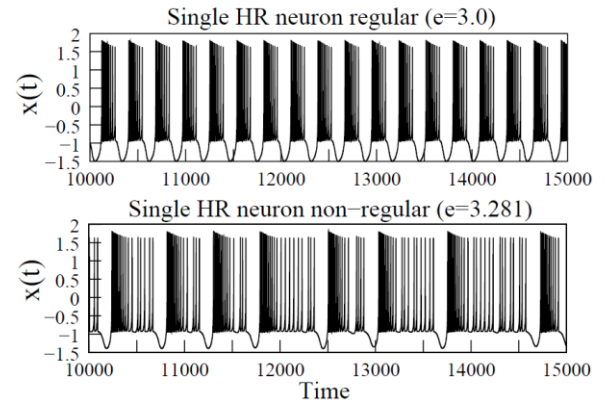
Regular activity:

$$e_i = 3.0, \mu = 0.0021 \text{ and } S = 4$$

Irregular activity:

$$e_i = 3.281$$

Reproduce the figures on the right



2- Regularization of the burst rhythm departing from the irregular activity by changing the injected current

Following the strategy described in this paper:

Closed-loop control of a minimal central pattern generator network. [Neurocomputing 170: 55-62.](#)

establish a measure of the regularization level and adaptively search for the minimal current level that leads to regular spiking-bursting activity.